

SHOP ZONE
Project Report Submitted
To

Gujarat University

In partial fulfilment of the requirements for
the award to the Degree of

5 YEAR INTEGRATED MASTER OF SCIENCE
(COMPUTER SCIENCE)
SEMESTER – V

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DEPARTMENT OF COMPUTER SCIENCE
GUJARAT UNIVERSITY, AHMEDABAD
YEAR: 2023-24

Department Of Computer Science

Gujarat University



Certificate

Roll No : 18

Seat No : 50035

This is to certify that Mr. /Ms. Mansuri Aasima Moahmmedimran student of Fifth Semester of 5 years Integrated M.Sc (Computer Science) has duly completed his/her project titled SHOP ZONE for the semester ending in December 2024, towards partial fulfillment of degree of 5 years Integrated M.Sc (Computer Science).

*30-12-2024
Date of Submission*

Internal Project Guide

Course Coordinator

Head of Department

Department Of Computer Science

Gujarat University



Certificate

Roll No :45

Seat No : 50083

This is to certify that Mr. /Ms Shah Khushi Urvishkumar student of Fifth Semester of 5 years Integrated M.Sc (Computer Science) has duly completed his/her project titled Shop Zone for the semester ending in December 2024, towards partial fulfillment of degree of 5 years Integrated M.Sc (Computer Science).

30-12-24

Date of Submission

Internal Project Guide

Course Coordinator

Head of Department

Department Of Computer Science

Gujarat University



Certificate

Roll No :46

Seat No : 50084

This is to certify that Mr. /Ms Shah Krisa D. student of Fifth Semester of 5 years Integrated M.Sc (Computer Science) has duly completed his/her project titled Shop Zone for the semester ending in December 2024, towards partial fulfillment of degree of 5 years Integrated M.Sc (Computer Science).

30-12-24

Date of Submission

Internal Project Guide

Course Coordinator

Head of Department

Department Of Computer Science
Gujarat University



Certificate

Roll No : 58

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30-12-2024
Date of Submission

Internal Project Guide

Course Coordinator

Head of Department

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About The Project

- **Project name : Shop Zone**

The project **Shop Zone** is an e-commerce platform designed to be a one-stop online shop for all fashion needs. The primary purpose of this platform is to offer high-quality apparels for men, women, and children while ensuring a convenient, secure, and user-friendly shopping experience.

The platform caters to both registered and non-registered users. Registered users include customers, admins, and delivery person, while non-registered users (guests) can browse products but need to register to make purchases.

- **Purpose and Objectives**

The purpose of **Shop Zone** is to bridge the gap between online shoppers and an efficient platform that provides the following:

1. Easy browsing and shopping experience.
2. Advanced filtering options for products based on categories, colors and prices,discount.
3. Easy order placement .

- **Key Features**

1. **User Registration and Login:**

New users can register by providing personal information like name, email, and password. Registered users can log in to access advanced functionalities such as managing their orders.

2. **Product Browsing and Filtering:**

Customers can explore apparel for men, women, and children. Filters allow sorting by color, discount, and price to help users find products that meet their choice.

3. **Product Details:**

Users can view detailed information about each product, including high-quality image, descriptions, price.

4. **Shopping Cart Management:**

Users can add selected products to their cart, edit quantities, or remove items before going to checkout.

5. **Order and Payment Management:**

Users can place orders and track their purchase history. The platform supports secure payment options, including cash on delivery.

- **Technologies**

The project is implemented using the following technologies:

- Frontend: HTML, CSS, Bootstrap (for responsive and user-friendly design).
- Backend: JSP, JDBC, and Servlets (for server-side logic).
- Database: MySQL (to store user, product, and order data).
- IDE : Eclipse
- Version Control : Github

- **Conclusion**

Shop Zone is a scalable platform aimed at simplifying online shopping for users. By offering essential features like product browsing, secure payments, and order management, it ensures a smooth and enjoyable shopping experience. The platform's focus on simplicity and functionality makes it a practical solution for online retail.

System Engineering

1. System Overview

The project **Shop Zone** is an online shopping platform designed to make shopping for clothes simple and convenient. It provides a variety of high-quality apparel for men, women, and children.

The platform allows users to:

- Search and browse products easily using filters like categories, colors, prices and discounts.
- Add items to a shopping cart and make secure payments.
- Manage their orders including tracking.

Different types of users can use the system:

- **Registered Users:** Customers, Admin, and Delivery Persons.
- **Guests:** Can only browse products but must register to make purchases.

2. Requirement Analysis and Feasibility Study

- Requirement Analysis

Key features of the system:

1. **User Registration:** Users can create accounts securely with their name, email, and password.
2. **Product Browsing:** Users can browse and search for products using filters like discount, color, and price.
3. **Product Details:** Users can view detailed information about each product, including images, description, price, and available sizes.
4. **Shopping Cart Management :** Users can add products to their shopping cart for later purchase. Users can edit quantities and remove items .
5. **Order Management:** Users can view past orders, track active orders, and cancel orders if needed.
6. **Payment Options:** Offers cash-on-delivery.

- Feasibility Study

This study check if the project is possible in different ways:

1. **Technical Feasibility:** The project can be built using common tools like HTML, CSS, JSP, and MySQL.
2. **Operational Feasibility:** The platform is easy to use and suits the operation performed by customers, admin, and delivery persons.
3. **Economic Feasibility:** Since it's an online platform, it reduces costs for setup and maintenance, making it affordable.

3. Proposed System

a. Objectives

The new system aims to:

1. Provide a simple and smooth shopping experience for customers.
2. Allow users to search and filter products easily.
3. Offer secure payments and order tracking.
4. Give admin tools to manage products, orders, and users effectively.

b. Hardware and Software Platforms

❖ Hardware Requirements

- **For Development and Deployment**
 - Processor: Intel Core i5 or better
 - RAM: 8 GB
 - Storage: 256 GB SSD or more
- **For Users (Client Systems):**
 - Processor: Intel Core i3 or better
 - RAM: 4 GB
 - Browser: Chrome

❖ Software Requirements

- **Development Tools:**

- **Frontend:**

- HTML(HyperText Markup Language)
 - CSS(Cascading Style Sheets)
 - Bootstrap

- **Backend:**

- JSP(JavaServer Pages.)
 - Servlets
 - JDBC(Java Database Connectivity.)

- **Database:**

- MySQL

- **Supported Operating Systems:**

- **Development:** Windows 10

- **Additional Tools:**

- **Code Sharing:** GitHub
 - **IDE:** Eclipse

System Design

1. System Architecture

- Flow of System

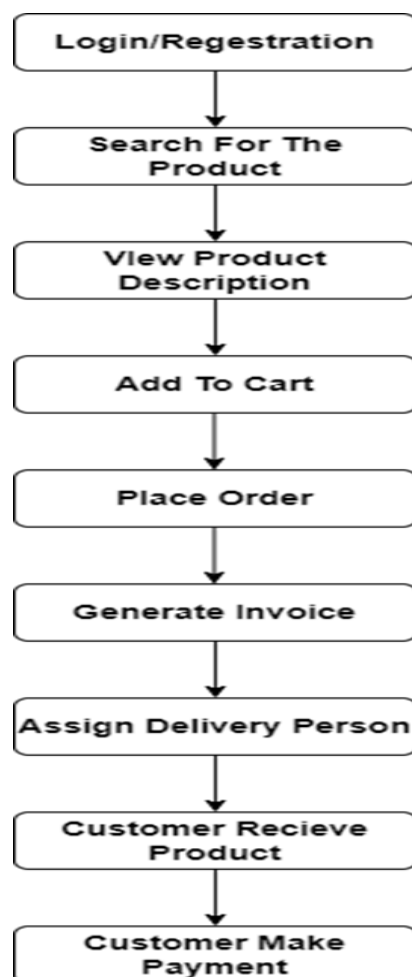


Figure 1 : Flow of system

- **Module hierarchy**



Figure 2 : Module hierarchy

- **ER - Diagram**

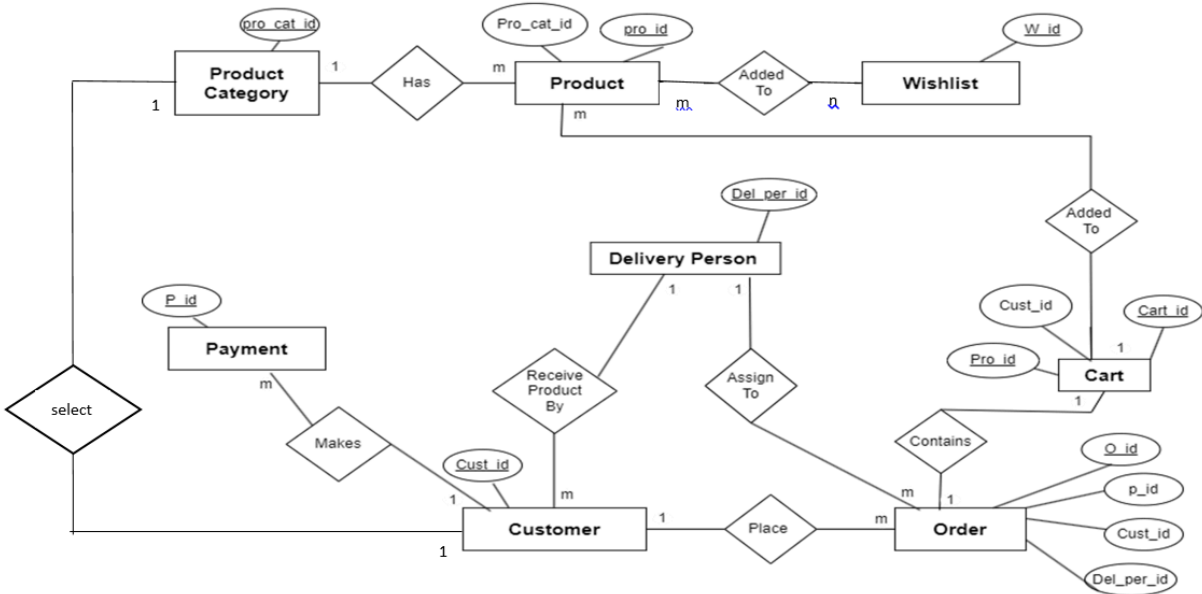


Figure 3 : ER - Diagram

- Use case diagrams

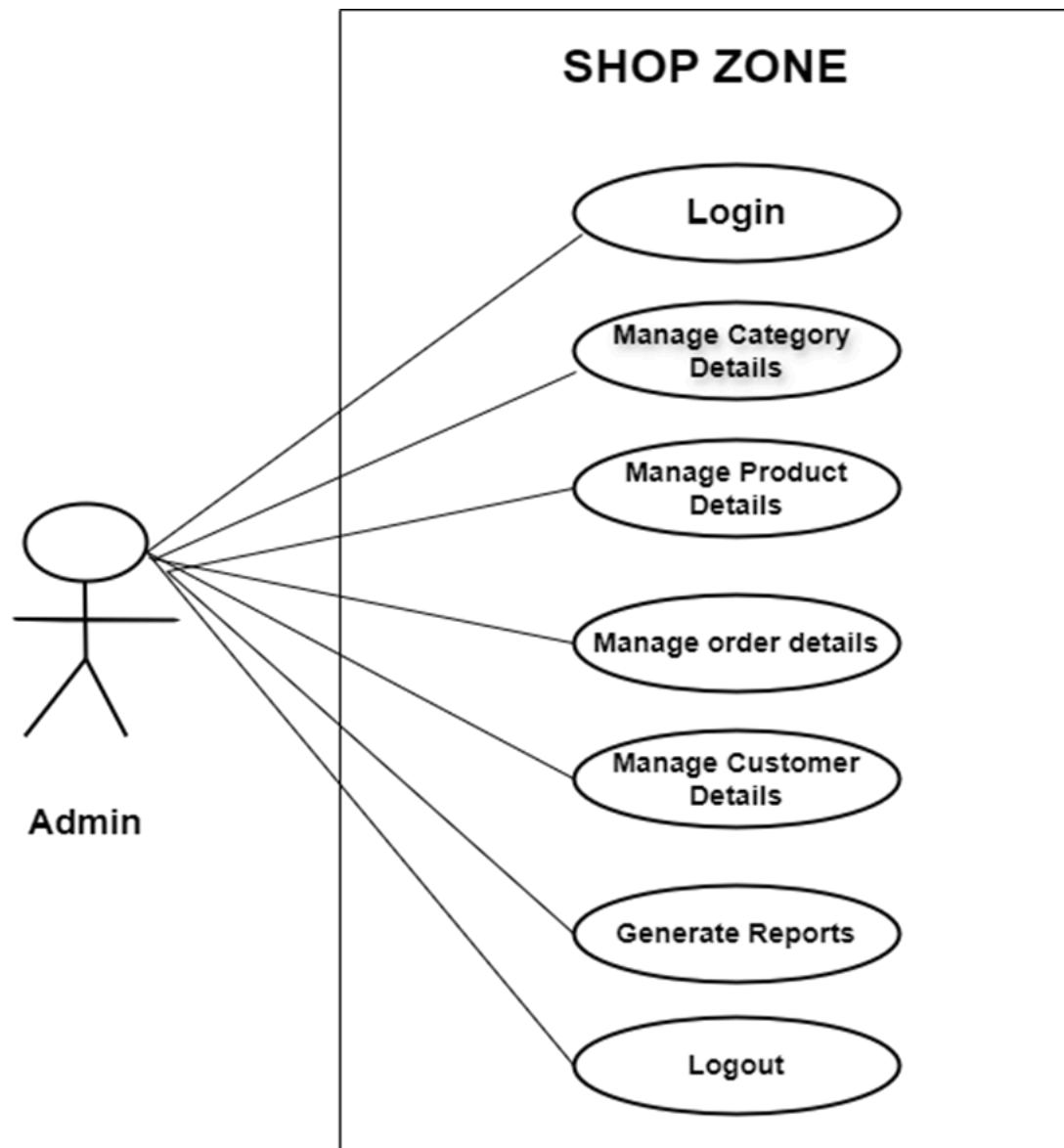


Figure 4 : use case diagram of admin

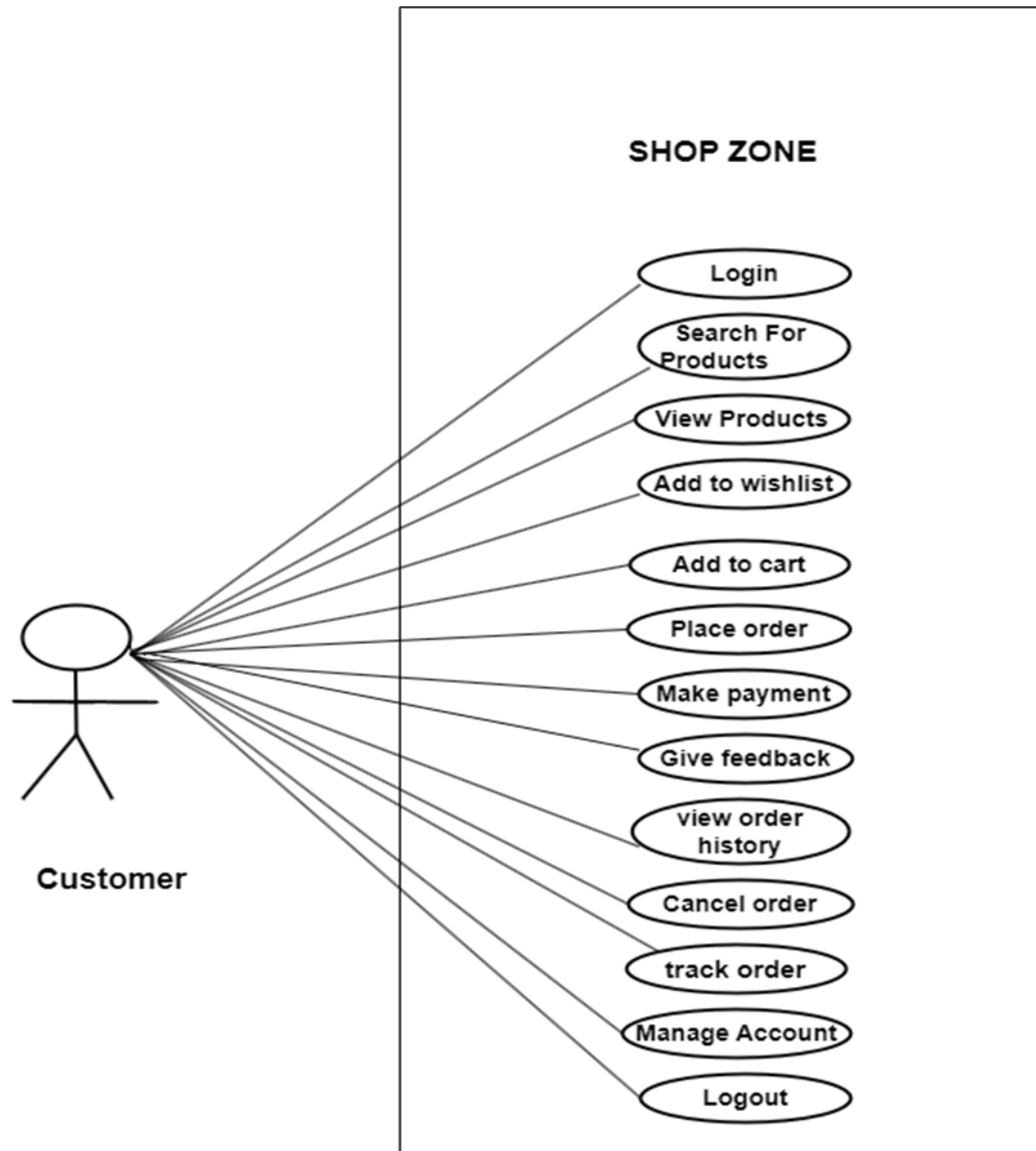


Figure 5 : use case diagram of customer

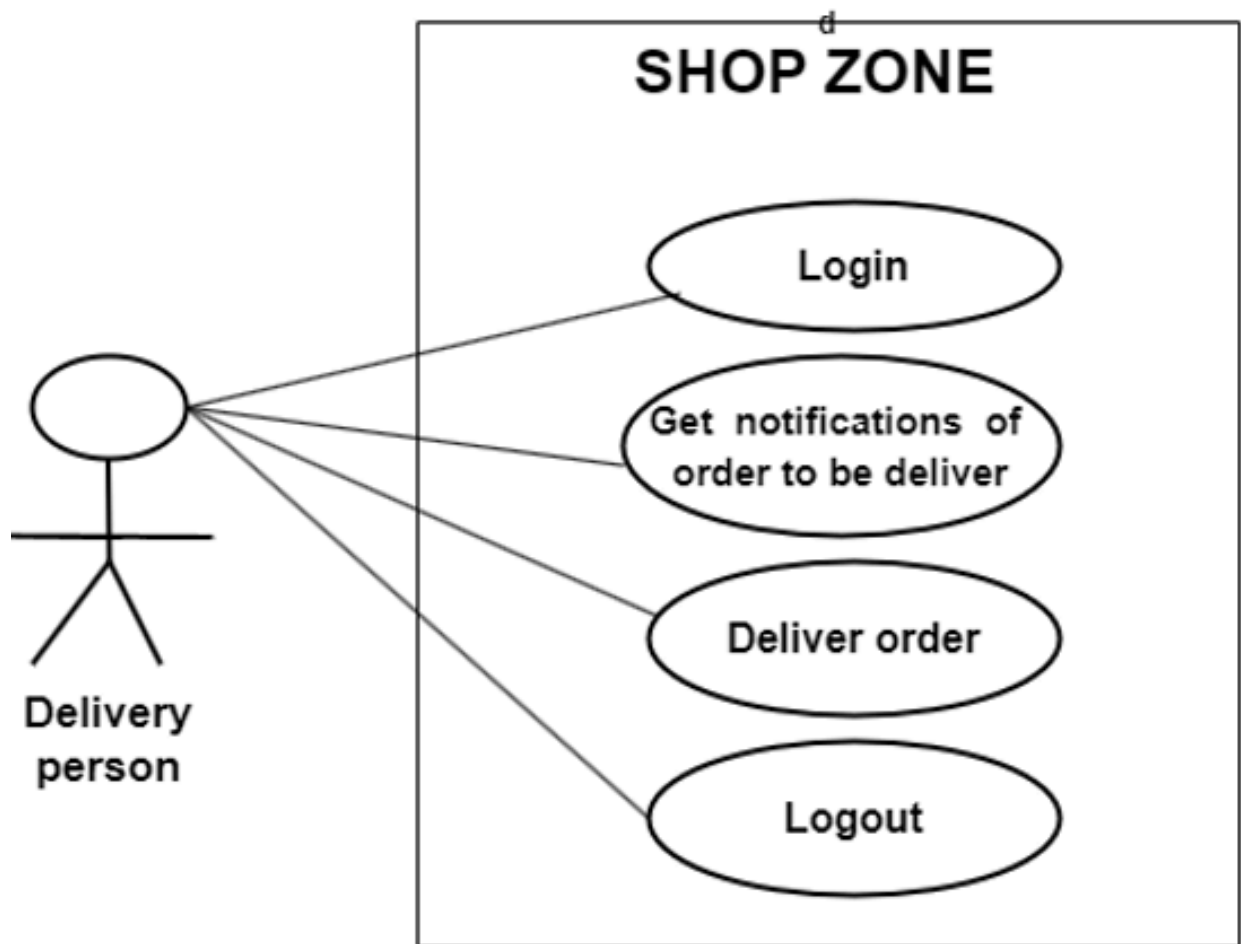


Figure 6 : use case diagram of delivery person

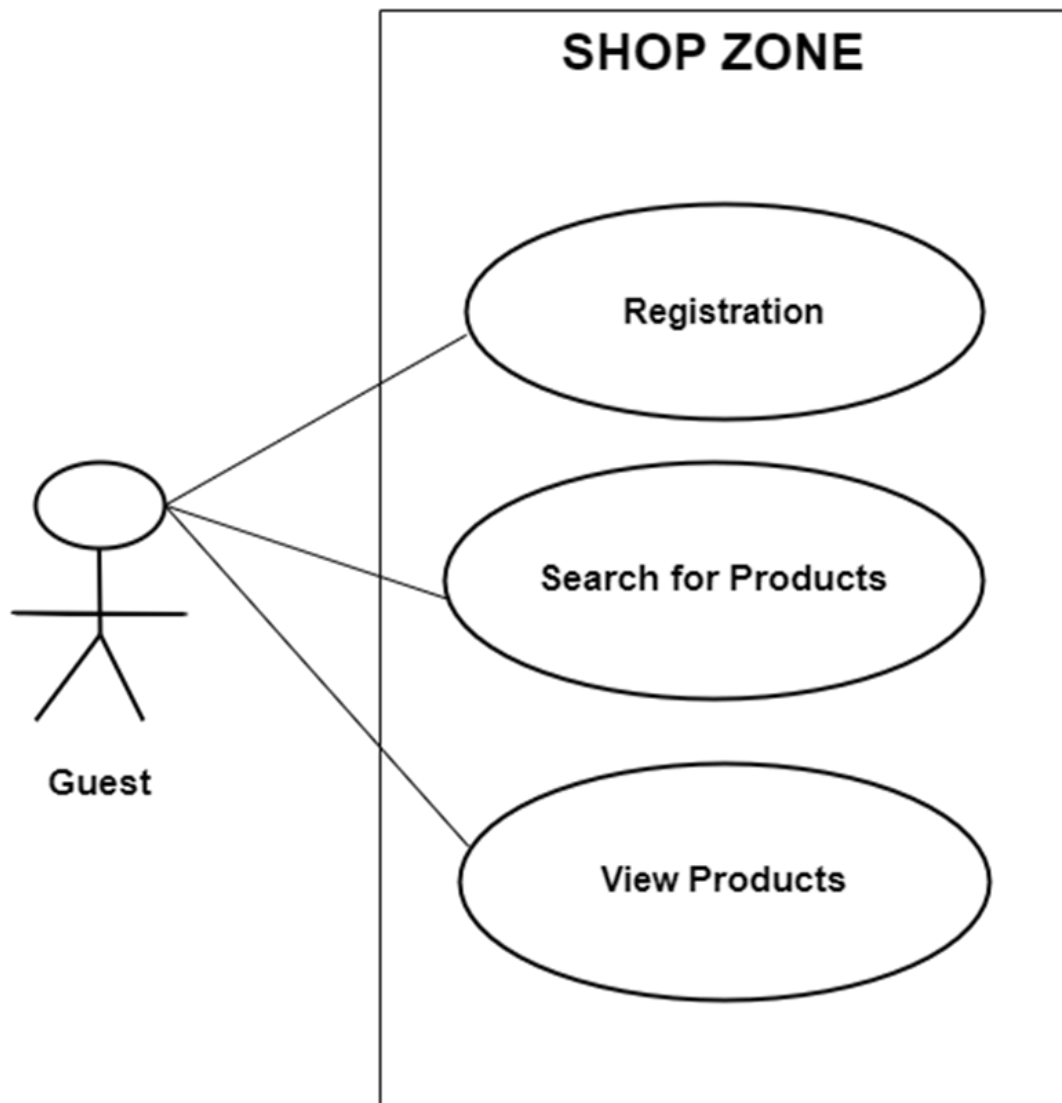


Figure 6 : use case diagram of guest

- **Activity diagrams**

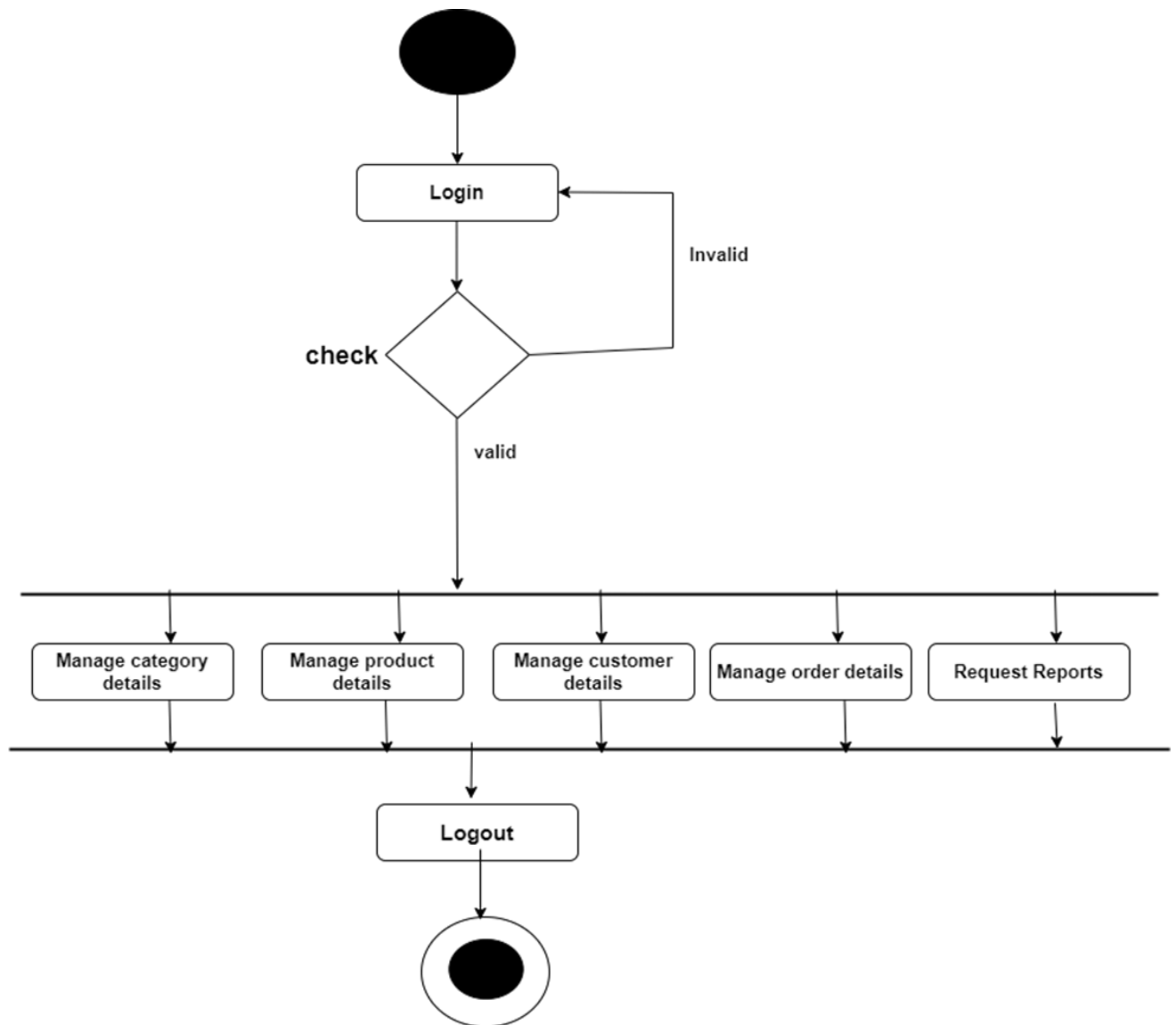


Figure 7 : Activity diagram of admin

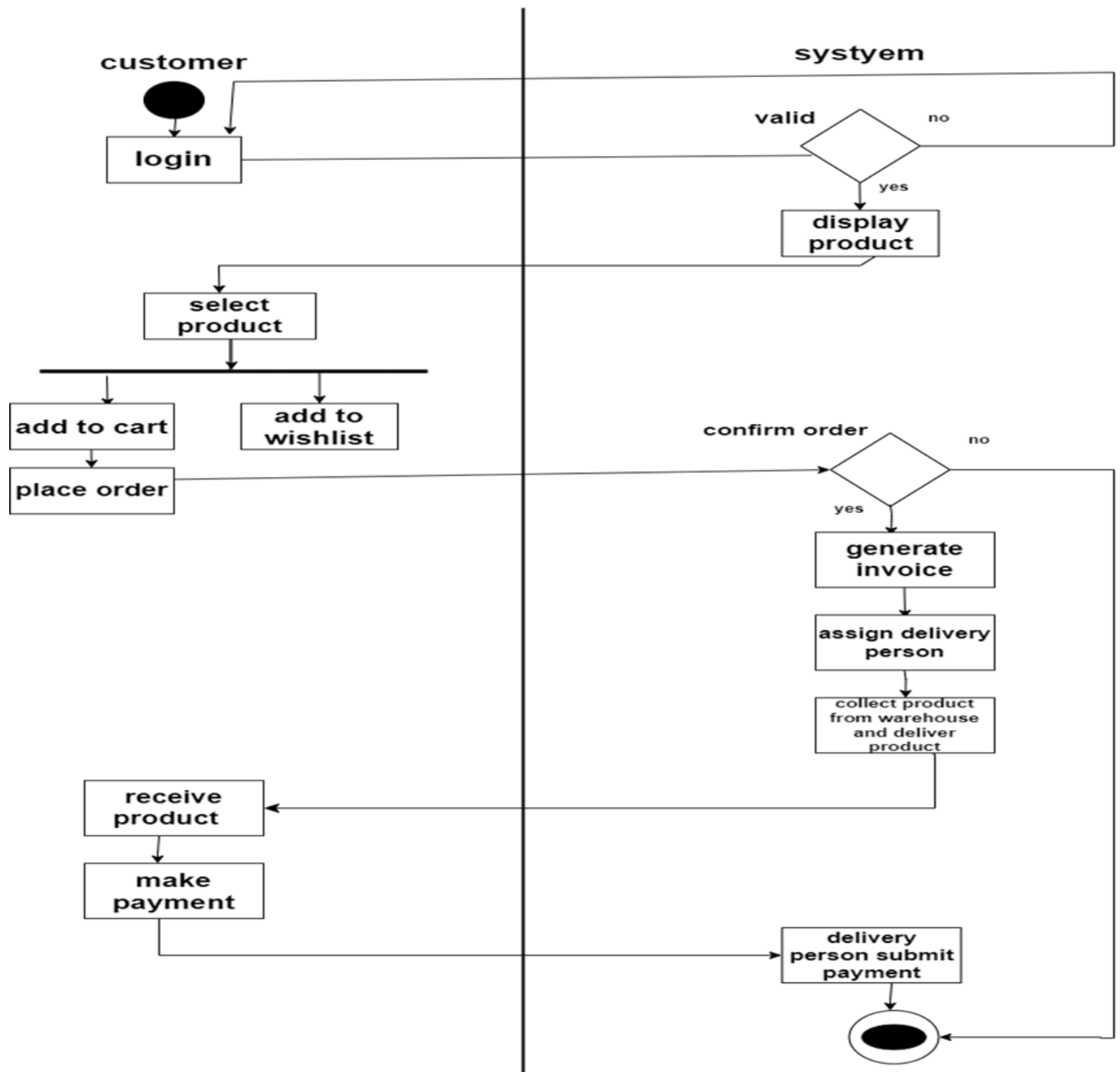


Figure 8 : Activity Diagram of User(customer , system)

2. Database Design

Product table

	Field	Type	Null	Key	Default	Extra
►	pro_id	int	NO	PRI	NULL	auto_increment
	title	varchar(255)	NO		NULL	
	description	varchar(255)	YES		NULL	
	sub_cat_id	int	YES	MUL	NULL	
	keywords	varchar(255)	YES		NULL	
	show_price	decimal(10,2)	YES		NULL	
	actual_price	decimal(10,2)	YES		NULL	
	stock	int	YES		NULL	
	pro_status	int	YES		NULL	
	reorder_limit	int	YES		NULL	
	discount	decimal(5,2)	YES		NULL	
	create_date	date	YES		NULL	
	update_date	date	YES		NULL	
	about	varchar(255)	NO		NULL	
	color	varchar(20)	YES		NULL	
	discounted...	decimal(10,2)	YES		NULL	

Customer table

	Field	Type	Null	Key	Default	Extra
►	c_id	int	NO	PRI	NULL	auto_increment
	username	varchar(100)	NO		NULL	
	email	varchar(100)	YES		NULL	
	default_address	varchar(255)	YES		NULL	
	password	varchar(255)	YES		NULL	
	phone_number	varchar(15)	YES		NULL	
	default_area	varchar(25)	YES		NULL	

Address Table

	Field	Type	Null	Key	Default	Extra
►	a_id	int	NO	PRI	NULL	auto_increment
	dp_id	int	YES	MUL	NULL	
	area_name	varchar(50)	YES		NULL	
	pin_code	varchar(8)	YES		NULL	

Cart Table

	Field	Type	Null	Key	Default	Extra
►	cart_id	int	NO	PRI	NULL	auto_increment
	pro_id	int	YES	MUL	NULL	
	c_id	int	YES	MUL	NULL	
	quantity	int	YES		NULL	

Delivery Person

	Field	Type	Null	Key	Default	Extra
►	dp_id	int	NO	PRI	NULL	auto_increment
	name	varchar(100)	YES		NULL	
	email	varchar(100)	YES		NULL	
	password	varchar(100)	YES		NULL	
	order_area1	varchar(100)	YES		NULL	
	order_area2	varchar(100)	YES		NULL	
	join_date	date	YES		NULL	
	total_delivery	int	YES		NULL	
	salary	decimal(10,2)	YES		NULL	

Employee Table

	Field	Type	Null	Key	Default	Extra
►	e_id	int	NO	PRI	NULL	auto_inc
	username	varchar(100)	YES		NULL	
	email	varchar(100)	YES		NULL	
	password	varchar(255)	YES		NULL	
	role	varchar(50)	YES		NULL	
	hire_date	date	YES		NULL	
	salary	decimal(10,2)	YES		NULL	

FeedBack Table

	Field	Type	Null	Key	Default	Extra
►	feedback_id	int	NO	PRI	NULL	auto_increment
	c_id	int	YES	MUL	NULL	
	pro_id	int	YES	MUL	NULL	
	content	varchar(255)	YES		NULL	
	rating	int	YES		NULL	

Image Table

	Field	Type	Null	Key	Default	Extra
►	img_id	int	NO	PRI	NULL	auto_increment
	path	varchar(255)	NO		NULL	
	pro_id	int	YES	MUL	NULL	
	created_at	timestamp	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED on update CURRENT_TI...

Order Line Item Table

	Field	Type	Null	Key	Default	Extra
►	order_line_item_id	int	NO	PRI	NULL	auto_increment
	order_id	int	YES	MUL	NULL	
	pro_id	int	YES	MUL	NULL	
	quantity	int	YES		NULL	
	prize	decimal(10,2)	YES		NULL	

Order Table

	Field	Type	Null	Key	Default	Extra
►	order_id	int	NO	PRI	NULL	auto_increment
	p_id	int	YES	MUL	NULL	
	c_id	int	YES	MUL	NULL	
	dp_id	int	YES	MUL	NULL	
	order_status	varchar(100)	YES		NULL	
	num_of_items	int	YES		NULL	
	order_date	date	YES		NULL	
	pincode	varchar(6)	YES		NULL	
	delivery_date	date	YES		NULL	
	shipping_address	varchar(255)	YES		NULL	

Parent Category Table

	Field	Type	Null	Key	Default	Extra
►	p_cat_id	int	NO	PRI	NULL	auto_increment
	p_cat_name	varchar(100)	NO		NULL	

Payment table

	Field	Type	Null	Key	Default	Extra
►	p_id	int	NO	PRI	NULL	auto_increment
	payment_method	varchar(50)	YES		NULL	
	payment_date	date	YES		NULL	
	amount	decimal(10,2)	YES		NULL	
	payment_status	varchar(50)	YES		NULL	
	payment_submit_date	datetime	YES		NULL	

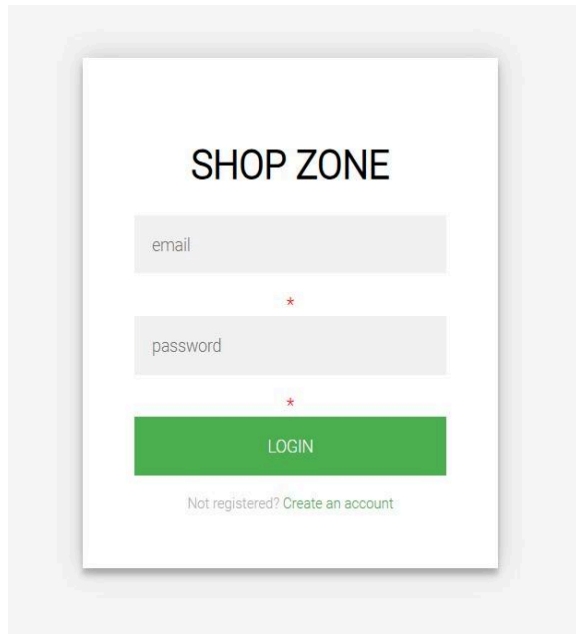
Sub Category

	Field	Type	Null	Key	Default	Extra
►	pro_id	int	NO	PRI	NULL	auto_increment
	title	varchar(255)	NO		NULL	
	description	varchar(255)	YES		NULL	
	sub_cat_id	int	YES	MUL	NULL	
	keywords	varchar(255)	YES		NULL	
	show_price	decimal(10,2)	YES		NULL	
	actual_price	decimal(10,2)	YES		NULL	
	stock	int	YES		NULL	
	pro_status	int	YES		NULL	
	reorder_limit	int	YES		NULL	
	discount	decimal(5,2)	YES		NULL	
	create_date	date	YES		NULL	
	update_date	date	YES		NULL	
	about	varchar(255)	NO		NULL	
	color	varchar(20)	YES		NULL	
	discounted...	decimal(10,2)	YES		NULL	

Whish List Table

	Field	Type	Null	Key	Default	Extra
►	w_id	int	NO	PRI	NULL	auto_increment
	c_id	int	YES	MUL	NULL	
	pro_id	int	YES	MUL	NULL	

3. Forms Design



SHOP ZONE

email

*

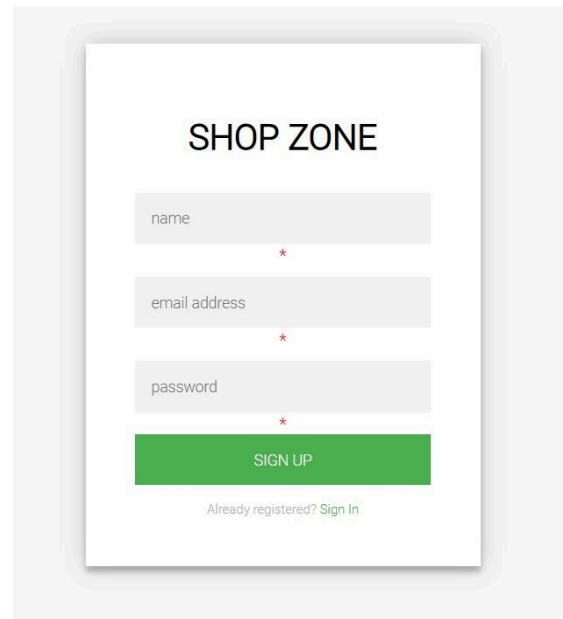
password

*

LOGIN

Not registered? [Create an account](#)

Sign in form



SHOP ZONE

name

*

email address

*

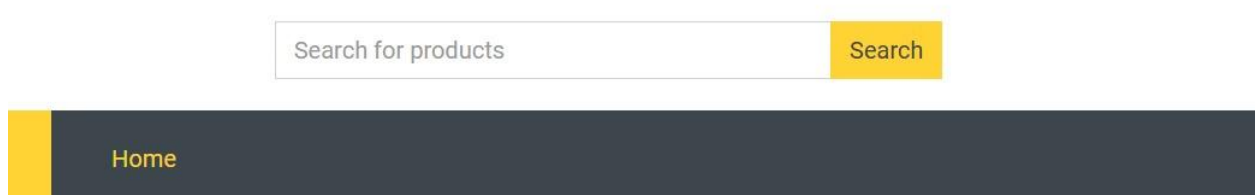
password

*

SIGN UP

Already registered? [Sign In](#)

sign up form



Search for products

Search

Home

Search Bar

Insert Subcategory

Subcategory Name

Select Parent Category

Subcategory Image

Insert SubCategory

Insert New Product

Title:

Description:

About:

Keywords:

Parent Category:

Sub Category:

Show Price:

Actual Price:

Stock:

Discount:

Image Path:

Insert Product

Insert Product

4. Reports Design

Sales Report

Start Date: 

End Date: 

Generate Report

product Sales Report

Start Date: 

End Date: 

Generate Report

5. System Security

1. Authentication and Authorization:

- Users must log in with a secure username and password.
- Role-based access control ensures that only authorized users (Customer, Admin, Delivery Person) can access specific features.

2. Data Security:

- Sensitive information like passwords is encrypted using secure hashing algorithms.

3. Error Handling and Input Validation:

- Proper error messages are displayed without exposing system details.
- User inputs are validated to prevent SQL injection and other attacks.

4. Session Management:

- Secure session cookies are used for user authentication.
- Automatic logout for inactive sessions helps prevent unauthorized access.

System Testing

System testing ensures that all components of the **Shop Zone** platform work together as expected. Different testing schemes were applied to validate the functionality, performance, and security of the system.

1. Testing Types

1. Functional Testing:

- Verified that all features, like user registration, login, product browsing, cart management, and order placement, work as expected.

2. Integration Testing:

- Ensured that different modules (e.g., user authentication, database) interact smoothly.

3. Usability Testing:

- Checked if the platform provides a user-friendly experience for customers, admin, and delivery person.

4. Security Testing:

- Validated that sensitive data, like user credentials and payment details, is encrypted and protected.

2. Test Case Data

Test Case ID	Description	Expected Result	Status
TC001	User Registration	User account created successfully	Passed
TC002	Add Product to Cart	Product added to the cart	Passed
TC003	Place order	Place order successfully	Passed
TC004	Search Functionality	Relevant results displayed	Passed
TC005	Admin Access Control	Admin-only actions restricted	Passed

3. Summary of Results

- All core functionalities of ShopZone were successfully tested.
- The system meets the requirements for functionality, usability, and security.
- Minor improvements were suggested to enhance performance and user experience.

Sample coding

BILLING ADDRESS

E-mail

krina@gmail.com

Address

123 Street

State

Gujarat

City

Ahmedabad

Search Area

Select Area

Total Amount: Rs. 7598.0

PAYMENT BY

☐ Cash on Delivery

Place Order

```

<!-- Checkout Page Start -->
<div class="container-fluid">
  <div class="row px-xl-5">

    <!-- Cart Details Section -->
    <div class="col-lg-7">
      <%
        // Fetch cart IDs from request parameter
        List<Integer> cartIds = new ArrayList<>();
        String cartIdsParam = request.getParameter("cart_ids");

        if (cartIdsParam != null && !cartIdsParam.isEmpty()) {
          String[] cartIdsArray = cartIdsParam.split(",");
          for (String id : cartIdsArray) {
            try {
              cartIds.add(Integer.parseInt(id.trim()));
            } catch (NumberFormatException e) {
              out.println("Invalid cart ID: " + id);
            }
          }
        }

        if (!cartIds.isEmpty()) {
          String cartIdCondition = String.join(",",
cartIds.stream().map(String::valueOf).toArray(String[]::new));
          Connection con = project.connection.getConnection();

          if (con != null) {
            String query = "SELECT c.cart_id, c.pro_id, p.title, p.show_price, p.discount,
c.quantity, " +
              "p.stock, (c.quantity * p.show_price) AS total, i.path " +
              "FROM cart c " +
              "JOIN product p ON c.pro_id = p.pro_id " +
              "JOIN image i ON i.pro_id = p.pro_id " +
              "WHERE c.cart_id IN (" + cartIdCondition + ")";

            PreparedStatement ps = con.prepareStatement(query);
            ResultSet rs = ps.executeQuery();

            if (!rs.next()) {
              out.println("<script>>window.location.href='cust_cart.jsp';</script>");
            } else {

```

```

grandTotal = 0;
total_num_items = 0;
%>

<!-- Cart Items Table -->
<table class="table table-light table-borderless table-hover text-center mb-0">
  <thead class="thead-dark">
    <tr>
      <th>Products</th>
      <th>Price</th>
      <th>Quantity</th>
      <th>Total</th>
    </tr>
  </thead>
  <tbody class="align-middle">
    <%
      do {
        String title = rs.getString("title");
        double price = rs.getDouble("show_price");
        double discount = rs.getDouble("discount");
        int quantity = rs.getInt("quantity");
        String imgPath = rs.getString("path");
        int discountPrice = (int) (price - ((price * discount) / 100));
        double total = quantity * discountPrice;
        grandTotal += total;
        total_num_items += quantity;
      %>
    <tr>
      <td class="align-middle">
        " style="width: 130px;
height: 150px;">
        <br><%= title %>
      </td>
      <td class="align-middle">Rs. <%= discountPrice %></td>
      <td class="align-middle"><%= quantity %></td>
      <td class="align-middle">Rs. <%= discountPrice * quantity %></td>
    </tr>
    <% } while (rs.next()); %>
  </tbody>
</table>
<%
}

```

```
        rs.close();
        ps.close();
    } else {
        out.println("Failed to connect to the database.");
    }
} else {
    out.println("No cart IDs provided.");
}
%>
</div>
<!-- Checkout Page End -->
```


Future Enhancements

1. Advanced Payment Options:

- additional payment methods like UPI, credit/debit cards, net banking, and digital wallets.

2. Real-Time Order Tracking:

- Providing real-time updates on the status of orders, including shipping and delivery details.

3. Discount Coupons and Loyalty Programs:

- Introducing discount codes and a loyalty rewards program to encourage repeat purchases.

4. Inventory Management for Admins:

- Adding features to help admins track stock levels and get alerts when inventory is low.

5. Customer care Support:

- Integrating chatbot to provide customer support and answer common queries.

Bibliography

Jsp and servlet :

- <https://docs.oracle.com/javaee/5/tutorial/doc/bnafe.html#:~:text=A%20servlet%20is%20a%20Java,a%20request%2Dresponse%20programming%20model>
- <https://www.oreilly.com/library/view/more-servlets-and/0130676144/>

Product images:

- Freepik
- Pixabay
- figma.com

